

Albert Poulose

Postdoctoral Researcher | Microgrid Research Center, Dept. of Electrical Engineering, Kyungpook National University, Daegu, South Korea

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Research profile

- Postdoctoral researcher in **power-system dynamic security** and **scientific machine learning**, focused on transient stability, converter-dominated grids (GFM/GFL), and physics-informed surrogates for trajectory prediction and critical clearing time (CCT) estimation.
- **Research areas:** Power-system stability | GFM/GFL converters | Scientific AI/PINNs | Digital twins
- **Key expertise:** Transient stability | Converter-dominated grids | Synchronizing torque & energy functions | PINNs | Parameter estimation | Python/PyTorch | MATLAB/Simulink | PSCAD

Selected research contributions

- **Physics-informed transient stability surrogate** (IEEE Access, 2026)
Developed a PINN-based trajectory surrogate for TSA and CCT estimation on an SMIB benchmark; foundation for multimachine and parameter-estimation extensions.
- **Converter transient stability** (IEEE Access, 2024)
Developed a transient-energy-function perspective for direct TSA of grid-connected voltage-source converters (GFM/GFL-relevant settings).
- **Synchronizing-torque stability metric** (Energies, 2022)
Proposed a synchronizing-torque-based transient stability index for multimachine interconnected systems.
- **Renewable-rich grid stability** (Energies, 2023)
Surveyed and structured transient stability analysis and enhancement techniques for renewable-rich power grids.

Research experience

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| 2025.03 to Present | <ul style="list-style-type: none">• Postdoctoral Researcher : Kyungpook National University, Daegu, South Korea
<i>Microgrid Research Center, Department of Electrical Engineering</i><ul style="list-style-type: none">• Developed physics-informed dynamic surrogates for TSA/CCT; extending to multimachine systems and parameter estimation.• Converter-dominated and low-inertia stability themes (GFM/GFL) with Python-centric dynamic simulation workflows.• First-author journal and conference outputs; collaboration with the Smart Power Systems Laboratory. |
| 2020.03 to 2025.02 | <ul style="list-style-type: none">• Graduate Research Assistant (Ph.D., full-time) : Kyungpook National University, Daegu, South Korea
<i>Smart Power Systems Lab, School of Electronic and Electrical Engineering</i>
Supervisor: Prof. Soobae Kim<ul style="list-style-type: none">• Thesis on multifaceted TSA: synchronizing torque in multimachine systems and energy-function methods for renewable-connected converters.• Lyapunov/TEF analysis, MATLAB/Simulink and Python simulation; four first-author journal papers spanning conventional and converter-dominated stability. |
| 2017.08 to 2019.06 | <ul style="list-style-type: none">• M.Tech research (Power Systems) : Government Engineering College, Thrissur (affiliated with APJ Abdul Kalam Technological University), India
Higher-order sliding-mode / super-twisting load frequency control for a two-area interconnected system (conference publications). |

Publications

4 journal articles | 3 conference papers | Google Scholar: 59 citations, h-index 4, i10-index 3 (as of Aug. 2026).

Journal Articles

- J1. A. Poulose and S. Kim, "Unified physics-informed trajectory surrogate for transient stability assessment and critical clearing time estimation on a single-machine infinite-bus benchmark," *IEEE Access*, vol. 14, pp. 122 177–122 196, 2026.  DOI: 10.1109/ACCESS.2026.3718419
- J2. A. Poulose and S. Kim, "Direct transient stability assessment of grid-connected voltage source converters: A transient energy functions perspective," *IEEE Access*, vol. 12, pp. 133 545–133 556, 2024.  DOI: 10.1109/ACCESS.2024.3450316

- J3. A. Poulou and S. Kim, "Transient stability analysis and enhancement techniques of renewable-rich power grids (review article)," *Energies*, vol. 16, no. 5, 2023, issn: 1996-1073.  DOI: 10.3390/en16052495
- J4. A. Poulou and S. Kim, "Synchronizing torque-based transient stability index of a multimachine interconnected power system," *Energies*, vol. 15, no. 9, 2022, issn: 1996-1073.  DOI: 10.3390/en15093432

Conference Proceedings

- C1. A. Poulou and S. Kim, "A technical briefing on machine learning for transient stability assessment in power systems," in *2025 International Conference on Advanced Information Technology (ADINTECH)*, Phnom Penh, Cambodia, 2025.  URL: <https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12333641>
- C2. A. Poulou and R. Kumar P., "Performance analysis of load frequency control for a two-area interconnected power system using twisting control," in *2019 International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2019.  URL: <https://sites.google.com/view/albertpoulou/publications?authuser=0>
- C3. A. Poulou and R. Kumar P., "Super-twisting algorithm based load frequency control of a two-area interconnected power system," in *2019 20th International Conference on Intelligent System Application to Power Systems (ISAP)*, New Delhi, India, 2019, pp. 1–5.  DOI: 10.1109/ISAP48318.2019.9065992

Research projects

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|--------------------|---|
| 2025.03 to Present | <ul style="list-style-type: none"> • Physics-Informed Dynamic Security Assessment
Physics-informed trajectory surrogates for TSA and CCT estimation; multimachine extensions; inertia/damping estimation and system identification. |
| 2022.03 to Present | <ul style="list-style-type: none"> • Converter-Dominated Grid Stability
GFM/GFL interactions, weak-grid and low-inertia stability, and transient-energy methods for renewable-rich converter-connected systems. |
| 2020.03 to 2025.02 | <ul style="list-style-type: none"> • Ph.D. research: Multifaceted transient stability analysis
Supervisor: Prof. Soobae Kim, Kyungpook National University
Synchronizing-torque-based stability indices in multimachine systems; Lyapunov and transient energy-function methods for renewable-connected grids with GFM/GFL converters. |

Current research directions

- **Physics-informed dynamic security:** Trajectory prediction, CCT estimation, and multimachine TSA with physics-informed surrogates.
- **Dynamic parameter estimation:** Inertia/damping estimation, system identification, and measurement-informed modeling.
- **Converter-dominated grids:** GFM/GFL dynamics, low-inertia systems, stability limits, and digital-twin applications.

Education

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| 2020.03 to 2025.02 | <ul style="list-style-type: none"> • Ph.D., Electronic and Electrical Engineering : Kyungpook National University, Daegu, South Korea
<i>School of Electronic and Electrical Engineering</i>
Thesis: Multifaceted Transient Stability Analysis of Power Systems: A Study of Synchronizing Torque in Conventional Systems and Energy Functions in Renewable Connected Systems
Advisor: Prof. Soobae Kim
GPA: 4.18/4.5 |
| 2017.08 to 2019.06 | <ul style="list-style-type: none"> • M.Tech., Power Systems : APJ Abdul Kalam Technological University, Kerala, India
<i>Government Engineering College, Thrissur</i>
GPA: 7.84/10
Thesis: Higher-order Sliding Mode Based Load Frequency Control for a Two-area Interconnected Power System |
| 2011.07 to 2015.05 | <ul style="list-style-type: none"> • B.Tech., Electrical and Electronics Engineering : Mahatma Gandhi University, Kerala, India
<i>SCMS School of Engineering and Technology, Ernakulam</i> |

Technical expertise

- **Research expertise:** Transient stability; small-signal stability; dynamic security; renewable integration; converter-dominated grids; low-inertia systems.
- **Converter dynamics:** GFM/GFL; VSM; PLL; droop control; weak-grid stability.

Technical expertise (continued)

- **Scientific AI:** PINNs; trajectory surrogates; hybrid physics-data modeling.
- **System identification & estimation:** Dynamic parameter estimation; inertia/damping estimation; measurement-informed modeling.
- **Simulation:** MATLAB/Simulink; PSCAD; PSS/E; PowerWorld; ANDES; PSAT; PST.
- **Programming:** Python; MATLAB; PyTorch; NumPy; SciPy; scikit-learn.

Languages • English (proficient), Malayalam (native), Korean (intermediate), Hindi (beginner), Tamil (beginner)

Research software and computational work

- Python/PyTorch physics-informed workflows for trajectory prediction, TSA screening, and CCT estimation; dynamic simulation pipelines for SMIB and multimachine systems; CCT computation and stability screening; ANDES- and MATLAB/Simulink-based modeling for conventional and converter-dominated grids; reproducible workflows for simulation, dataset construction, and surrogate training.

Selected teaching

- University teaching concurrent with research: Logic Circuits (69 students) and Numerical Analysis (26 students) as Lecturer at KNU (2025–2026); Linear Control Systems and Electric Hybrid Vehicles as Assistant Professor at CCET (2019); UG TA support during M.Tech. Mentoring of junior researchers within the laboratory.

Awards and achievements

- 2020.03 • **KNU International Graduate Scholarships (KINGS):** Tuition fee waiver for two years of Ph.D. coursework.
- 2016.03 • **Graduate Aptitude Test in Engineering (GATE):** Qualified GATE-2016 (score 411; 90.90th percentile); M.Tech. stipend via AICTE (India).

References

- Ph.D. Supervisor • **Prof. Soobae Kim**
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- M.Tech. Supervisor • **Dr. Ramesh Kumar P.**, Associate Professor
Government Engineering College, Kozhikode, Kerala, India
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